CUTOFF FOR POLYMER PINNING IN THE ENTROPICALLY REPULSIVE PHASE

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Abstract. We consider the heat-bath corner-flip dynamics for nonnegative nearest-neighbor paths of even length L, pinned at zero at both endpoints, with equilibrium weight $\lambda^{\mathcal{N}}$, where $\mathcal{N}$ counts the interior contacts with the wall. For every fixed $\lambda \in (1,2)$, we prove worst-case total variation cutoff at $L^2 \log L/\pi^2$. This removes the restriction to extremal initial conditions in the previously known upper bound for this regime. The proof uses a quantitative compensation inequality for the wall terms to control the expected integral of the positive drift of the area between coupled paths. A lower bound on the infinitesimal variance, combined with a concave Lyapunov function, then yields coalescence in an additional time of order L^2.

1. Introduction

The polymer considered here is a discrete path constrained to stay above a hard wall. It evolves by local corner flips, and each contact with the wall receives a weight λ. We study the time needed for this dynamics to approach equilibrium from the most unfavorable initial path. The regime $1 < \lambda < 2$ is particularly delicate. The wall attracts the polymer energetically, but it also repels it entropically, because it forbids every excursion below it. In this regime the entropic repulsion prevails, and the equilibrium polymer is delocalized. We refer to $\lambda \in (1,2)$ as the *entropically repulsive phase*. We prove worst-case cutoff throughout this phase; together with [Yan21], this completes the cutoff picture for fixed pinning parameters in the delocalized phase.

Let $L \geq 4$ be even. We consider the state space

$$\Omega_L = \left\{ \xi \in \mathbb{Z}^{L+1} : \ \xi_0 = \xi_L = 0; \ \ \xi_x \geq 0, \ 0 \leq x \leq L; \ |\xi_{x+1} - \xi_x| = 1, \ 0 \leq x < L \ \right\}. \tag{1.1}$$

For $\lambda \geq 0$, the polymer pinning measure on Ω_L is defined by

$$\mathcal{N}(\xi) = \sum_{x=1}^{L-1} \mathbf{1}_{\{\xi_x = 0\}}, \qquad \mu_L^{\lambda}(\xi) = \frac{\lambda^{\mathcal{N}(\xi)}}{Z_L(\lambda)}. \tag{1.2}$$

Here $Z_L(\lambda) = \sum_{\xi \in \Omega_L} \lambda^{\mathcal{N}(\xi)}$, with the convention $0^0 = 1$.

Entropic repulsion of the hard wall. The parameter λ is the reward of a contact with the wall, so the interaction is energetically repulsive for $\lambda < 1$ and attractive for $\lambda > 1$. Without the positivity constraint, that is, for bridges of simple random walk that may visit negative heights, every attraction localizes the polymer: the critical value is $\lambda_c = 1$. The hard wall shifts the critical value to $\lambda_c = 2$; see [Fis84], [Gia07, Chapter 2] and [CMT08, Section 1]. The shift is entropic. Between two consecutive contacts, a path in Ω_L performs an excursion that must stay strictly above the wall, and these excursions carry only half of the mass of the excursions of simple random walk:

$$\sum_{m \geq 1} 2^{-2m} E_{2m} = \frac{1}{2}, \tag{1.3}$$

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where E_{2m} is the number of strictly positive excursions of length $2m$, that is, the Catalan number C_{m-1}. More precisely, if $0 = x_0 < x_1 < \cdots < x_k = L$ and $n_i = x_i - x_{i-1}$, then

$$\sum_{\substack{\xi\in\Omega_L:\\ \{x:\xi_x=0\}=\{x_0,\dots,x_k\}}} 2^{-L}\lambda^{\mathcal{N}(\xi)} = \frac{1}{\lambda}\prod_{i=1}^{k}\lambda\, 2^{-n_i}E_{n_i}.$$

The contacts thus form a renewal structure with inter-arrival weights $\lambda 2^{-n}E_n$, whose total mass is $\lambda/2$; without the wall, it would be λ. In other words, each contact gains the energetic factor λ but costs the entropic factor $1/2$. For $\lambda < 2$ the polymer is delocalized: the expected number of contacts is bounded uniformly in L, and typical heights are of order $\sqrt{L}$. For $\lambda > 2$ it is localized, with of order L contacts [CMT08, Section 1]. In the entropically repulsive phase $\lambda \in (1, 2)$, the wall attracts the polymer, but the entropic repulsion due to the hard wall dominates.

For $\lambda > 0$, let $\mathcal{L} = \mathcal{L}_{L,\lambda}$ be the generator of the continuous-time single-site heat-bath dynamics:

$$(\mathcal{L}f)(\xi) = \sum_{x=1}^{L-1}\left[\mu_L^\lambda(f \mid (\xi_y)_{y\neq x}) - f(\xi)\right]. \tag{1.4}$$

Each site is thus resampled at rate one. When $\xi_{x-1} = \xi_{x+1} \geq 1$, a change of the height at x replaces ξ_x by $\xi_{x-1} + \xi_{x+1} - \xi_x$. The rate of this corner flip is

$$R_x^\lambda(\xi) = \begin{cases} 1/2, & \xi_{x-1} = \xi_{x+1} \geq 2,\\ 1/(1+\lambda), & (\xi_{x-1},\xi_x,\xi_{x+1}) = (1,0,1),\\ \lambda/(1+\lambda), & (\xi_{x-1},\xi_x,\xi_{x+1}) = (1,2,1),\\ 0, & \text{otherwise.}\end{cases} \tag{1.5}$$

We refer to Figure 1 for a graphical explanation. The chain is irreducible and reversible with respect to μ_L^λ.

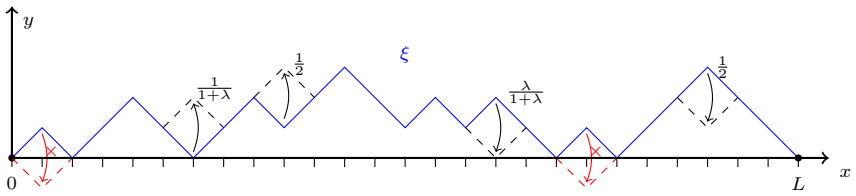


FIGURE 1. A graphical representation of the jump rates for the system pinned at $(0, 0)$ and $(L, 0)$. A transition of the dynamics corresponds to flipping a corner, whose rate depends on how it changes the number of contact points with the x-axis. The rates are chosen in a manner such that the dynamics is reversible with respect to μ_L^λ. The two red dashed corners are not available and labeled with $\times$, because of the nonnegative restriction of the state space Ω_L. Note that not all the possible transitions are shown in the figure.

Let $\sigma_t^{\xi,\lambda}$ denote the chain started from ξ, and let $P_t^{\xi,\lambda}$ be its law at time t. We suppress λ in the notation for the process when it is fixed. For $\lambda > 0$, define

$$\begin{aligned} d^{L,\lambda}(t) &= \max_{\xi\in\Omega_L}\|P_t^{\xi,\lambda} - \mu_L^\lambda\|_{\mathrm{TV}},\\ T_{\mathrm{mix}}^{L,\lambda}(\varepsilon) &= \inf\{t \geq 0 : d^{L,\lambda}(t) \leq \varepsilon\}, \qquad \varepsilon \in (0,1),\end{aligned} \tag{1.6}$$

where $\|\nu - \nu'\|_{\mathrm{TV}} = \frac{1}{2}\sum_{\xi\in\Omega_L}|\nu(\xi) - \nu'(\xi)|$ is the total variation distance between ν, ν'.

The dynamics was introduced by Caputo, Martinelli and Toninelli [CMT08], who proved mixing-time bounds of order $L^2 \log L$ in the delocalized phase. In [Yan21, Theorem 1.1], it identified the leading constant $1/\pi^2$ and established worst-case cutoff for $\lambda \in [0, 1]$. For $\lambda \in (1, 2)$,

in [Yan21, Theorem 1.2] it treats the maximum of the mixing times from the two extremal paths

$$\wedge_x = \min(x, L-x), \qquad \vee_x = x - 2\lfloor x/2 \rfloor, \qquad 0 \le x \le L, \tag{1.7}$$

under the coordinatewise order. In particular, the distance

$$d^{L,\lambda}_{\mathrm{ext}}(t) = \max_{\xi \in \{\wedge,\vee\}} \|P^{\xi,\lambda}_t - \mu^{\lambda}_L\|_{\mathrm{TV}}$$

has cutoff at $L^2 \log L/\pi^2$. The present paper extends the upper bound to the maximum over all initial paths.

Theorem 1.1. *For every fixed $\lambda \in (1,2)$ and $\varepsilon \in (0,1)$, we have*

$$\lim_{\substack{L\to\infty \\ L \text{ even}}} \frac{\pi^2 T^{L,\lambda}_{\mathrm{mix}}(\varepsilon)}{L^2 \log L} = 1.$$

Equivalently, the sequence of chains exhibits total variation cutoff at

$$t_L = \frac{L^2 \log L}{\pi^2}, \tag{1.8}$$

for every fixed $\delta \in (0,1)$,

$$d^{L,\lambda}((1-\delta)t_L) \longrightarrow 1, \qquad d^{L,\lambda}((1+\delta)t_L) \longrightarrow 0.$$

The matching lower bound is already contained in [Yan21, Proposition 3.1]; our contribution is the upper bound.

Comparison with [Yan21]. The analysis of [CMT08, Yan21] rests on an exact formula for the drift of each height [Yan21, Lemma 2.3], which goes back to [CMT08, Lemma 2.3]: for every $\xi \in \Omega_L$ and $1 \le x \le L-1$,

$$\mathcal{L}\xi_x = \frac{\xi_{x-1}+\xi_{x+1}}{2} - \xi_x + \mathbf{1}_{\{\xi_{x-1}=\xi_{x+1}=0\}} + \kappa\, \mathbf{1}_{\{\xi_{x-1}=\xi_{x+1}=1\}}, \qquad \kappa = \frac{1-\lambda}{1+\lambda}, \tag{1.9}$$

where $\mathcal{L}\xi_x$ denotes $\mathcal{L}f(\xi)$ for $f(\xi) = \xi_x$. The two indicators record the two effects of the wall. The first one is the entropic repulsion of the hard wall: between two contacts, the height at x cannot decrease, and its update is forced to equal one. This term does not depend on λ. The second one is the energetic interaction: a corner whose two neighbors are at height one falls onto the wall with probability $\lambda/(1+\lambda)$, so this term pulls the path toward the wall exactly when $\lambda > 1$.

For $\lambda \in [0,1]$ we have $\kappa \ge 0$, and both wall terms in (1.9) are nonnegative and decreasing in the path. This allows the term-by-term comparison of [Yan21, Section 4]: if $\xi' \le \xi$, then at every site the wall terms of ξ are at most those of ξ' [Yan21, (4.11)]. Consequently, for the weighted area $\Phi_\beta(\xi) = \sum_{x=1}^{L-1} \cos(\beta(x-L/2)/L)\,\xi_x$, with $\beta \in (0,\pi)$ and $\gamma_\beta = 1-\cos(\beta/L)$, the generator contracts the area between any two ordered paths pointwise [Yan21, (4.12)]:

$$(\mathcal{L}\Phi_\beta)(\xi) - (\mathcal{L}\Phi_\beta)(\xi') \le -\gamma_\beta(\Phi_\beta(\xi) - \Phi_\beta(\xi')), \qquad \xi' \le \xi. \tag{1.10}$$

The area between two coupled ordered paths is then a supermartingale, and in [Yan21, Section 4] the supermartingale method of [LL20, Section 7] yields their coalescence. For $\lambda \in (1,2)$, however, $\kappa < 0$: the attractive term is increasing in the path, and the term-by-term comparison fails. If both neighbors of a site are at height one in the lower path but not in the upper path, the attraction pulls the lower path toward the wall and increases the area between the two paths. Thus (1.10) fails, and the area is not a supermartingale [Yan21, Section 5]. To bypass this obstruction, [Yan21, Section 5] applies the censoring inequality of Peres and Winkler [PW13] to the dynamics started from the maximal and the minimal paths. This inequality is available only for these two initial paths, and it controls the laws of the corresponding chains rather than their coalescence. This is the origin of the restriction to extremal initial paths in [Yan21, Theorem 1.2].

Novelty. We remove this restriction by replacing the term-by-term comparison with a comparison in expectation, in which the entropic repulsion of the hard wall compensates the energetic

attraction. Lemma 2.3 shows that, under every probability measure with a monotone density with respect to μ_L^λ, the deviation of the attractive term from its equilibrium mean is at most a fraction $1-\theta$ of the deviation of the hard-wall term, where

$$\theta = \left(\frac{2-\lambda}{\lambda}\right)^2 \in (0,1).$$

This is a strict, quantitative form of the wall compensation in [CMT08, Section 4.1]. Its proof is local: conditional FKG compares, at each site, the configurations that activate the attractive term with those that activate the hard-wall term at the two neighboring sites. The comparison coefficient is then bounded below by means of the partition-function asymptotics of the delocalized phase. This is where the entropic repulsion enters quantitatively; in particular, $\theta \downarrow 0$ as $\lambda \uparrow 2$. The laws of the chains started from $\wedge$ and $\vee$ have monotone densities at every deterministic time. Hence, at every such time, the expected adverse drift of the area is at most $1-\theta$ times its expected favorable drift from the hard wall. The margin $\theta > 0$ then bounds the expected integral of the adverse drift by a constant multiple of the mean area (Proposition 2.4). Lemma 2.3 applies to the laws of the chains at deterministic times, not to conditional laws given the past, so the area is still not a supermartingale. We therefore replace the supermartingale argument of [Yan21, Section 4], which relies on stopping times, by an absorption estimate for a concave Lyapunov function (Lemma 4.1) that uses estimates at deterministic times only.

Proof strategy. We couple the chains $(X_t^+)_{t\ge0}$ and $(X_t^-)_{t\ge0}$ started from $\wedge$ and $\vee$ respectively with a stationary chain $(Z_t)_{t\ge0}$, so that $X_t^- \le Z_t \le X_t^+$ for all $t \ge 0$. The coupling uses independent Poisson clocks indexed by site, height, and direction, as in [LL20, Appendix A]. It suffices to show that both ordered pairs (X^+, Z) and (Z, X^-) coalesce. For either pair, let A be the difference between the weighted areas Φ_β of the upper and the lower path. By (1.9), its drift decomposes as

$$b = -\gamma_\beta A - r + j, \qquad \gamma_\beta \sim \frac{\beta^2}{2L^2},$$

where $r \ge 0$ comes from the hard-wall term and decreases the area, while $j \ge 0$ comes from the attractive term and increases it. The compensation inequality gives $\mathbb{E}j_t \le (1-\theta)\mathbb{E}r_t$ at every deterministic time t, and in Section 2 we deduce

$$\mathbb{E}A_T \le \frac{L^2}{4}e^{-\gamma_\beta T}, \qquad \mathbb{E}\int_T^\infty j_t\,\mathrm{d}t \le \frac{1-\theta}{\theta}\mathbb{E}A_T. \tag{1.11}$$

The balance between initial area L^2 and equilibrium area $L^{3/2}$ explains the time scale $L^2\log L/\beta^2$. At a slightly later time T, the mean area is at most $L^{3/2-a}$ for some $a > 0$.

Organization of the paper. Section 3 supplies the geometric input for the remaining coalescence. At each deterministic time after $(1+\alpha)t_L$, with fixed $\alpha > 0$, the upper path has height at most $\sqrt{L}\log L$ and the stationary path has no monotone stretch longer than $\lceil(\log L)^2\rceil$, with probability tending to one. The height estimate uses the known convergence from the maximal initial path in [Yan21, Theorem 1.2]. On these regular states, counting active corners gives the infinitesimal variance bound

$$v \ge c(1 + A/K_L), \qquad K_L = \sqrt{L}\log L\,\lceil(\log L)^2\rceil.$$

In Section 4, an increasing concave function with $F''(a) = -K_L/(K_L+a)$ converts this variance bound into a negative drift of constant order. The positive drift is controlled by the second estimate in (1.11). The resulting absorption bound gives

$$\mathbb{P}(A_{T+L^2} > 0) \le C\frac{\sqrt{L}(\log L)^4}{L^2}\mathbb{E}A_T + o(1) = o(1).$$

Only estimates at deterministic times are needed; no event of uniform regularity over the entire interval is used. Finally, letting $\beta \uparrow \pi$ recovers the sharp constant.

Throughout the rest of the paper, $\lambda \in (1,2)$ is fixed and L tends to infinity through even integers. All logarithms are natural. Constants may depend on λ and on fixed auxiliary parameters, but not on L.

2. Monotone coupling and wall compensation

We write $\xi \le \xi'$ if $\xi_x \le \xi'_x$ for every x. For probability measures ν, ν' on Ω_L, the notation $\nu \preceq \nu'$ means that $\nu(f) \le \nu'(f)$ for every increasing function f, where $\nu(f) = \sum_{\xi\in\Omega_L} f(\xi)\nu(\xi)$.

2.1. **A monotone coupling with independent height clocks.** The coupling must preserve order and also allow paths at different heights to move independently. The first property reduces mixing to coalescence; the second will give the variance bound in Section 3.

For each $x \in \{1, \dots, L-1\}$ and each attainable height h, let $\mathcal{P}^{x,h,\uparrow}$ and $\mathcal{P}^{x,h,\downarrow}$ be independent Poisson processes with respective rates

$$r_\uparrow(h) = \begin{cases} (1+\lambda)^{-1}, & h = 0, \\ 1/2, & h \ge 1, \end{cases} \qquad r_\downarrow(h) = \begin{cases} \lambda/(1+\lambda), & h = 2, \\ 1/2, & h \ge 3. \end{cases}$$

The downward processes at heights zero and one are omitted. At a point of $\mathcal{P}^{x,h,\uparrow}$, every coupled path whose height at x is h and whose two neighbors have height $h+1$ is raised by two at x. A point of $\mathcal{P}^{x,h,\downarrow}$ acts analogously on a peak of height h. All other coordinates remain unchanged. By (1.5), every marginal has generator $\mathcal{L}$.

The coupling preserves order. Indeed, if $\xi \le \xi'$ and a clock raises $\xi_x = h$, either $\xi'_x \ge h+2$, or $\xi'_x = h$ and the nearest-neighbor constraint forces both neighbors of ξ' to be $h+1$, so both paths flip. The downward case is identical. Equal paths therefore remain equal.

Let $X_t^+ = \sigma_t^\wedge$, $X_t^- = \sigma_t^\vee$, and $Z_t = \sigma_t^\mu$, where $Z_0 \sim \mu_L^\lambda$ independently of the clocks. We write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation under this coupling. The triple (X_t^+, Z_t, X_t^-) is a finite-state Markov chain, and Z_t has law μ_L^λ at every time. Then

$$X_t^- \le Z_t \le X_t^+, \qquad d^{L,\lambda}(t) \le \mathbb{P}(X_t^+ \ne Z_t) + \mathbb{P}(Z_t \ne X_t^-). \tag{2.1}$$

The second inequality follows by including an arbitrary initial path in the same grand coupling.

2.2. **Weights and wall terms.** We use the positive cosine weights of [Yan21, Section 4]. These weights have a simple discrete Laplacian, which isolates the linear relaxation term in the area drift. We take $\beta \in (0, \pi)$ to keeps every weight bounded away from zero, including near the endpoints; this will also give a lower bound on every nonzero area jump. Fix $\beta \in (0, \pi)$ and set

$$w_x = \cos\left(\frac{\beta(x - L/2)}{L}\right), \quad c_\beta = \cos(\beta/2) > 0, \quad \Phi_\beta(\xi) = \sum_{x=1}^{L-1} w_x \xi_x, \quad \gamma_\beta = 1 - \cos(\beta/L).$$

Thus $c_\beta \le w_x \le 1$ and $(w_{x-1} + w_{x+1})/2 = \cos(\beta/L) w_x \le w_x$. Define

$$I_x^0(\xi) = \mathbf{1}_{\{\xi_{x-1}=\xi_{x+1}=0\}}, \qquad I_x^1(\xi) = \mathbf{1}_{\{\xi_{x-1}=\xi_{x+1}=1\}}, \qquad R_i(\xi) = \sum_{x=1}^{L-1} w_x I_x^i(\xi).$$

Both R_0 and R_1 are decreasing in the path order. Indeed, $\xi_x \equiv x \pmod 2$, so each nonzero indicator above specifies the smallest possible values of its two neighboring coordinates. By the same parity remark, $I_x^0 \equiv 0$ for even x and $I_x^1 \equiv 0$ for odd x. Recall from (1.9) that $\kappa = (1-\lambda)/(1+\lambda)$; it is negative since $\lambda > 1$.

Lemma 2.1 (Weighted-area drift)**.** *For every $\xi \in \Omega_L$, we have*

$$\mathcal{L}\Phi_\beta = -\gamma_\beta \Phi_\beta - c_\beta + R_0 + \kappa R_1. \tag{2.2}$$

Proof. Write $a = \xi_{x-1}$ and $b = \xi_{x+1}$. If $a \ne b$, or $a = b \ge 2$, the heat-bath update at x has mean $(a+b)/2$. If $a = b = 0$, the update is forced to equal $1 = (a+b)/2 + 1$. If $a = b = 1$, it equals 0 with probability $\lambda/(1+\lambda)$ and 2 with probability $1/(1+\lambda)$, so its mean is $2/(1+\lambda) = 1 + \kappa$. Hence the conditional mean of the update at x is

$$\frac{\xi_{x-1} + \xi_{x+1}}{2} + I_x^0(\xi) + \kappa I_x^1(\xi).$$

Multiplying the resulting formula for $\mathcal{L}\xi_x$ by w_x and summing by parts gives

$$\sum_{x=1}^{L-1} w_x \left(\frac{\xi_{x-1}+\xi_{x+1}}{2} - \xi_x \right) = -\gamma_\beta \Phi_\beta(\xi) - \frac{w_0\xi_1 + w_L\xi_{L-1}}{2}.$$

Since $w_0 = w_L = c_\beta$ and $\xi_1 = \xi_{L-1} = 1$, the boundary term equals $-c_\beta$. □

2.3. **An elementary partition-function bound.** We record a uniform-in-length estimate used twice below: first to control local contact probabilities in the compensation argument, and later to transfer a tail bound from μ_L^1 to μ_L^λ.

Lemma 2.2 ([CMT08, Theorem 2.1])**.** *For each fixed $\alpha \in [0,2)$ there are positive constants a_α, A_α such that [Delocalized partition function]*

$$a_\alpha 2^n n^{-3/2} \le Z_n(\alpha) \le A_\alpha 2^n n^{-3/2}, \qquad n \ge 2 \text{ even.} \tag{2.3}$$

2.4. **Compensation under monotone densities.** The following conditional FKG comparison gives a quantitative form of the wall compensation in [CMT08, Section 4.1]. Its role is to show that the favorable contribution from R_0 exceeds the adverse contribution from R_1 by a fixed fraction. This is an inequality for expectations under monotone densities; no pointwise comparison of the two wall terms is required. Set

$$\theta = \left(\frac{2-\lambda}{\lambda} \right)^2 \in (0,1).$$

Lemma 2.3 (Compensation of the wall terms)**.** *If ν has an increasing density with respect to μ_L^λ, then*

$$\Delta_i := \mu_L^\lambda(R_i) - \nu(R_i) \ge 0, \qquad |\kappa|\Delta_1 \le (1-\theta)\Delta_0.$$

If ν has a decreasing density, the same statement holds with $\Delta_i := \nu(R_i) - \mu_L^\lambda(R_i)$.

Proof. The measure μ_L^λ satisfies the FKG lattice condition: taking the pointwise maximum and minimum of two paths preserves the sum of their contact counts. More explicitly, for any two paths ξ, ξ',

$$\mu_L^\lambda(\min(\xi,\xi'))\mu_L^\lambda(\max(\xi,\xi')) = \mu_L^\lambda(\xi)\mu_L^\lambda(\xi'),$$

where minimum and maximum are taken coordinatewise. Conditioning on specified values of coordinates restricts the measure to a sublattice and preserves this identity. Thus every such conditional measure is positively associated. Write $\nu = h\mu_L^\lambda$ with h increasing. FKG gives $\nu \succeq \mu_L^\lambda$, and hence $\Delta_i \ge 0$. For the same reason,

$$\delta_x := \mu_L^\lambda(I_x^0) - \nu(I_x^0) \ge 0, \qquad 1 \le x \le L-1,$$

and $\delta_x = 0$ for even x.

Local comparison. For even $y \in \{2, \dots, L-2\}$ let

$$B_y = \{\xi_{y-1} = \xi_{y+1} = 1\}, \quad C_y^- = \{\xi_{y-2} = \xi_y = 0\}, \quad C_y^+ = \{\xi_y = \xi_{y+2} = 0\}.$$

Note that $\mathbf{1}_{B_y} = I_y^1$, $\mathbf{1}_{C_y^-} = I_{y-1}^0$ and $\mathbf{1}_{C_y^+} = I_{y+1}^0$. Each $C_y^\pm$ is the intersection of B_y with a decreasing event D. Under $\mu_L^\lambda(\cdot \mid B_y)$, which satisfies FKG, the density of $\nu(\cdot \mid B_y)$ is proportional to h and hence increasing, so $\nu(D \mid B_y) \le \mu_L^\lambda(D \mid B_y)$ whenever $\nu(B_y) > 0$. Therefore

$$\nu(C_y^\pm) \le a_y^\pm \nu(B_y), \qquad a_y^\pm = \frac{\mu_L^\lambda(C_y^\pm)}{\mu_L^\lambda(B_y)},$$

with the same inequality holding trivially if $\nu(B_y) = 0$. Consequently,

$$\mu_L^\lambda(C_y^\pm) - \nu(C_y^\pm) \ge a_y^\pm(\mu_L^\lambda(B_y) - \nu(B_y)). \tag{2.4}$$

A uniform lower bound on the comparison coefficient. Under $\mu_L^\lambda(\cdot \mid B_y)$ the weight factorizes over $[0, y-1]$, $\{y\}$ and $[y+1, L]$, so $\xi_y \in \{0,2\}$ is independent of the rest of the path and equals

0 with probability $\lambda/(1+\lambda)$. Given $\xi_y = 0$, the path splits at the contact y into independent pinned paths of lengths y and $L-y$. This shows that

$$a_y^- = \frac{\lambda}{1+\lambda} p_y, \qquad a_y^+ = \frac{\lambda}{1+\lambda} p_{L-y},$$

where $p_2 = 1$ and, for even $n \geq 4$, $p_n = \mu_n^\lambda(\xi_2 = 0) = \lambda Z_{n-2}(\lambda)/Z_n(\lambda)$ (we used reflection symmetry for a_y^-). By FKG for the decreasing events $\{\xi_2 = 0\}$ and $\{\xi_n = 0\}$ under μ_{n+2}^λ,

$$p_{n+2} \leq \mu_{n+2}^\lambda(\xi_2 = 0 \mid \xi_n = 0) = p_n.$$

Moreover, (2.3) implies $Z_{2m}(\lambda)^{1/m} \to 4$. The ratios $r_m = Z_{2m}(\lambda)/Z_{2m-2}(\lambda) = \lambda/p_{2m}$, $m \geq 2$, are nondecreasing. Since $\prod_{k=2}^m r_k = Z_{2m}(\lambda)$ and $Z_2(\lambda) = 1$, their geometric means converge to 4. A nondecreasing positive sequence has the same limit as its geometric means, so $r_m \uparrow 4$. Hence $p_n \downarrow \lambda/4$, and

$$a_y^\pm \geq \frac{\lambda^2}{4(1+\lambda)}, \qquad \frac{|\kappa|}{a_y^\pm} \leq \frac{4(\lambda-1)}{\lambda^2} = 1-\theta. \tag{2.5}$$

Summation with the cosine weights. By (2.4), (2.5) and $\delta_x \geq 0$,

$$|\kappa|(\mu_L^\lambda(B_y) - \nu(B_y)) \leq \frac{|\kappa|}{a_y^\pm}\,\delta_{y\pm1} \leq (1-\theta)\delta_{y\pm1}.$$

Average the two inequalities, multiply by w_y, and sum over even y. Since $R_1 = \sum_{y \text{ even}} w_y \mathbf{1}_{B_y}$, this gives

$$\begin{aligned}|\kappa|\Delta_1 &\leq \frac{1-\theta}{2} \sum_{\substack{2\leq y\leq L-2\\ y \text{ even}}} w_y(\delta_{y-1} + \delta_{y+1}) \\ &\leq (1-\theta)\sum_{x=1}^{L-1} \frac{w_{x-1}+w_{x+1}}{2}\delta_x \leq (1-\theta)\Delta_0.\end{aligned}$$

The middle inequality includes the additional nonnegative endpoint terms; the last inequality uses $(w_{x-1}+w_{x+1})/2 \leq w_x$.

For decreasing h, FKG gives $\nu \preceq \mu_L^\lambda$ and hence $\delta_x := \nu(I_x^0) - \mu_L^\lambda(I_x^0) \geq 0$. Since B_y is decreasing, $\nu(B_y) \geq \mu_L^\lambda(B_y) > 0$. Conditional FKG now gives $\nu(C_y^\pm) \geq a_y^\pm \nu(B_y)$, and therefore

$$\nu(C_y^\pm) - \mu_L^\lambda(C_y^\pm) \geq a_y^\pm(\nu(B_y) - \mu_L^\lambda(B_y)).$$

The same weighted summation proves the assertion. □

2.5. **Mean-area decay and accumulated positive drift.** We now apply compensation to the two coupled pairs. The resulting proposition is the only input about the wall interaction needed in the coalescence argument.

Define the two nonnegative areas

$$A_t^+ = \Phi_\beta(X_t^+) - \Phi_\beta(Z_t), \qquad A_t^- = \Phi_\beta(Z_t) - \Phi_\beta(X_t^-).$$

For either sign, write A_t for the corresponding area and set

$$\begin{aligned} r_t^+ &= R_0(Z_t) - R_0(X_t^+), & j_t^+ &= |\kappa|(R_1(Z_t) - R_1(X_t^+)), \\ r_t^- &= R_0(X_t^-) - R_0(Z_t), & j_t^- &= |\kappa|(R_1(X_t^-) - R_1(Z_t)).\end{aligned}$$

All four quantities are nonnegative. In this notation, r reduces the area and j increases it. In the common filtration,

$$A_t = A_0 + M_t + \int_0^t b_u\,\mathrm{d}u, \qquad b_t = -\gamma_\beta A_t - r_t + j_t \leq j_t. \tag{2.6}$$

By Dynkin's formula and Lemma 2.1, M^+ is the difference of the martingale parts of $\Phi_\beta(X^+)$ and $\Phi_\beta(Z)$, while M^- is the corresponding difference for $\Phi_\beta(Z)$ and $\Phi_\beta(X^-)$. These are martingales in the common filtration because each marginal has the prescribed generator in that filtration.

Proposition 2.4 (Area decay and integrated drift)**.** *For either sign and every deterministic* $T \geq 0$*,*

$$\mathbb{E}A_T \leq \frac{L^2}{4} e^{-\gamma_\beta T}, \qquad \mathbb{E}\int_T^\infty j_t \, \mathrm{d}t \leq \frac{1-\theta}{\theta} \mathbb{E}A_T. \tag{2.7}$$

Proof. Without loss of generality, we focus on A^+. Reversibility and monotonicity imply that the law of X_t^+ has increasing density with respect to μ_L^λ: that density at ξ is $P_t(\xi, \wedge)/\mu_L^\lambda(\wedge)$, an increasing function of ξ because $\{\wedge\}$ is an increasing event. The density for X_t^- is decreasing. Since $Z_t \sim \mu_L^\lambda$, Lemma 2.3 gives, for both signs,

$$\mathbb{E}j_t \leq (1-\theta)\mathbb{E}r_t.$$

Thus $u(t) := \mathbb{E}A_t$, which is differentiable because the chain is finite, satisfies

$$u'(t) = -\gamma_\beta u(t) - \mathbb{E}r_t + \mathbb{E}j_t \leq -\gamma_\beta u(t) - \theta \mathbb{E}r_t. \tag{2.8}$$

Since $0 \leq u(0) \leq \Phi_\beta(\wedge) \leq L^2/4$ and $r_t \geq 0$, Gronwall's inequality gives the first bound. Integrating (2.8) from T to S and using $u \geq 0$ gives $\theta \int_T^S \mathbb{E}r_t \, \mathrm{d}t \leq u(T)$. Using $\mathbb{E}j_t \leq (1-\theta)\mathbb{E}r_t$ yields

$$\int_T^S \mathbb{E}j_t \, \mathrm{d}t \leq \frac{1-\theta}{\theta} u(T).$$

By Tonelli's theorem and monotone convergence theorem as $S \to \infty$, we prove the second bound. □

3. Fluctuations and regularity

The preceding section controls the mean area and its positive drift. To make the remaining area hit zero, we also need a lower bound on its fluctuations. The geometric reason for such a bound is simple: if two paths differ on many sites, and one path has no long stretch of constant slope, then many corners can flip in just one of the two paths. The height-clock coupling makes each such flip contribute to the variance of the area.

3.1. **Variance from active corners.** Let $\zeta \in \Omega_L$. A site $x \in \{1, \ldots, L-1\}$ is a *valley* of ζ if $\zeta_{x-1} = \zeta_{x+1} = \zeta_x + 1$, and a *peak* of ζ if $\zeta_{x-1} = \zeta_{x+1} = \zeta_x - 1$. The edge between $y-1$ and y is *ascending* if $\zeta_y - \zeta_{y-1} = 1$ and *descending* if $\zeta_y - \zeta_{y-1} = -1$. Let

$$Q(\zeta) = \max\{n \geq 1 : \ |\zeta_{i+n} - \zeta_i| = n \text{ for some } i \in \{0, \ldots, L-n\}\} \tag{3.1}$$

be the maximal number of consecutive edges of ζ having the same slope.

Update maps. To define the infinitesimal variance of the area, we describe the coupling of Section 2.1 by update maps. Let $\mathcal{C}$ be the finite set of labels $c = (x, h, \epsilon)$ of the Poisson processes $\mathcal{P}^c = \mathcal{P}^{x,h,\epsilon}$ introduced there. For $c = (x, h, \epsilon) \in \mathcal{C}$ we write $x_c = x$; we let $s_c = 1$ if $\epsilon = \uparrow$ and $s_c = -1$ if $\epsilon = \downarrow$; and we let r_c be the rate of $\mathcal{P}^c$, that is, $r_c = r_\uparrow(h)$ or $r_c = r_\downarrow(h)$ according as $\epsilon = \uparrow$ or $\epsilon = \downarrow$. For $\zeta \in \Omega_L$ define

$$\iota_c(\zeta) = \mathbf{1}_{\{\zeta_x = h, \ \zeta_{x-1} = \zeta_{x+1} = h + s_c\}}, \qquad (T_c\zeta)_y = \zeta_y + 2 s_c \, \iota_c(\zeta) \mathbf{1}_{\{y = x\}} \quad (0 \leq y \leq L). \tag{3.2}$$

Thus $\iota_c(\zeta) = 1$ exactly when x is a valley of ζ at height h (if $\epsilon = \uparrow$) or a peak of ζ at height h (if $\epsilon = \downarrow$). In that case a point of $\mathcal{P}^c$ flips the corner of ζ at x, and $T_c\zeta$ is the resulting path; otherwise $T_c\zeta = \zeta$. By construction, at each point of $\mathcal{P}^c$ every coupled path ζ is replaced by $T_c\zeta$, and $T_c\xi \geq T_c\xi'$ whenever $\xi \geq \xi'$, as shown in Section 2.1. Hence a coupled pair (ξ, ξ'), consisting of an upper and a lower path, is a Markov chain on $\{(\xi, \xi') \in \Omega_L^2 : \xi \geq \xi'\}$ with generator

$$\mathcal{L}^{(2)} f(\xi, \xi') = \sum_{c \in \mathcal{C}} r_c \big[f(T_c\xi, T_c\xi') - f(\xi, \xi') \big]. \tag{3.3}$$

Definition of v. For $\xi \geq \xi'$ in Ω_L set

$$A(\xi, \xi') = \Phi_\beta(\xi) - \Phi_\beta(\xi'), \qquad H(\xi) = \max_{0 \leq x \leq L} \xi_x.$$

The infinitesimal variance of the area at (ξ,ξ') is

$$v(\xi,\xi') = \sum_{c\in\mathcal{C}} r_c\big[A(T_c\xi,T_c\xi') - A(\xi,\xi')\big]^2 = \big(\mathcal{L}^{(2)}A^2 - 2A\,\mathcal{L}^{(2)}A\big)(\xi,\xi'). \tag{3.4}$$

Since $\Phi_\beta(T_c\zeta) - \Phi_\beta(\zeta) = 2s_c w_{x_c}\iota_c(\zeta)$, the clock c changes the area by

$$A(T_c\xi,T_c\xi') - A(\xi,\xi') = 2s_c w_{x_c}\big(\iota_c(\xi) - \iota_c(\xi')\big). \tag{3.5}$$

This increment vanishes unless c acts on exactly one of the two paths. We call such clocks *active* for the pair and set

$$\mathcal{C}^*(\xi,\xi') = \{c\in\mathcal{C} :\ \iota_c(\xi)\neq\iota_c(\xi')\}.$$

Since $(\iota_c(\xi)-\iota_c(\xi'))^2$ is the indicator of $\{c\in\mathcal{C}^*(\xi,\xi')\}$, (3.4) and (3.5) give the explicit formula

$$v(\xi,\xi') = 4\sum_{c\in\mathcal{C}^*(\xi,\xi')} r_c\, w_{x_c}^2. \tag{3.6}$$

Relation with the coupled triple. For $y=(\xi,\zeta,\xi')$ with $\xi'\le\zeta\le\xi$, put $A^+(y)=A(\xi,\zeta)$ and $A^-(y)=A(\zeta,\xi')$. Then $A_t^\pm = A^\pm(Y_t)$ for the coupled triple $Y_t=(X_t^+,Z_t,X_t^-)$, and we set

$$v_t^+ = v(X_t^+,Z_t), \qquad v_t^- = v(Z_t,X_t^-).$$

As for A_t, we write v_t for either of them. The triple jumps from y to $T_cy=(T_c\xi,T_c\zeta,T_c\xi')$ at rate r_c, so its transition rates are $q(y,z)=\sum_{c\in\mathcal{C}:\,T_cy=z} r_c$ for $z\neq y$. Since $A^+(y)$ does not depend on the third coordinate of y,

$$\sum_{z\neq y} q(y,z)\big(A^+(z)-A^+(y)\big)^2 = \sum_{c\in\mathcal{C}} r_c\big(A^+(T_cy)-A^+(y)\big)^2 = v(\xi,\zeta), \tag{3.7}$$

and in the same way the left-hand side with A^- in place of A^+ equals $v(\zeta,\xi')$. In words, transitions that move only the third path do not change the area, and v_t is the quantity $v(Y_t)$ of Lemma 4.1 for the observable $A=A^\pm$. Moreover, by [KL99, Appendix 1, Lemma 5.1], the process $(M_t^\pm)^2 - \int_0^t v_u^\pm\,\mathrm{d}u$ is a martingale, where $M^\pm$ is the martingale in (2.6). Thus v_t is the density of the predictable quadratic variation of M: $\langle M^\pm\rangle_t = \int_0^t v_u^\pm\,\mathrm{d}u$.

The next lemma is the analogue, for the present coupling, of [Yan21, Lemma 4.7]; compare also [LL20, Lemma 24]. It bounds v from below by exhibiting active clocks at corners lying in the discrepancy set

$$S(\xi,\xi') = \{x\in\{0,\dots,L\} :\ \xi_x\neq\xi'_x\}.$$

Lemma 3.1 (Variance from active corners)**.** *Let $\xi\ge\xi'$ be distinct elements of Ω_L, and write $A=A(\xi,\xi')$, $H=H(\xi)$ and $S=S(\xi,\xi')$. For either choice $\zeta\in\{\xi,\xi'\}$,*

$$\#\mathcal{C}^*(\xi,\xi') \ge \max\Big(1,\frac{|S|}{2Q(\zeta)}\Big) \qquad \text{and} \qquad |S|\ge\frac{A}{H}. \tag{3.8}$$

Consequently,

$$v(\xi,\xi') \ge c_{\lambda,\beta}\Big(1+\frac{A}{HQ(\zeta)}\Big), \qquad c_{\lambda,\beta}=\frac{c_\beta^2}{1+\lambda}. \tag{3.9}$$

Moreover, for every $c\in\mathcal{C}$, the increment $A(T_c\xi,T_c\xi')-A(\xi,\xi')$ is either zero or has absolute value in $[2c_\beta,2]$.

Here A/H is a lower bound on the number $|S|$ of discrepancy sites, and division by $Q(\zeta)$ accounts for the possible spacing of the corners of ζ. The additional constant in (3.9) is essential when only a few discrepancies remain; it comes from the downward clock that every component of S carries (Step 3 below). Figure 2 illustrates the proof.

Proof. Fix $\zeta\in\{\xi,\xi'\}$, and let $\tilde\zeta$ denote the other member of the pair.

Step 1: reduction to counting active clocks. Since $\lambda>1$, every rate satisfies

$$r_c \ge \min\Big\{\frac12,\frac{1}{1+\lambda},\frac{\lambda}{1+\lambda}\Big\} = \frac{1}{1+\lambda}, \qquad c\in\mathcal{C}.$$

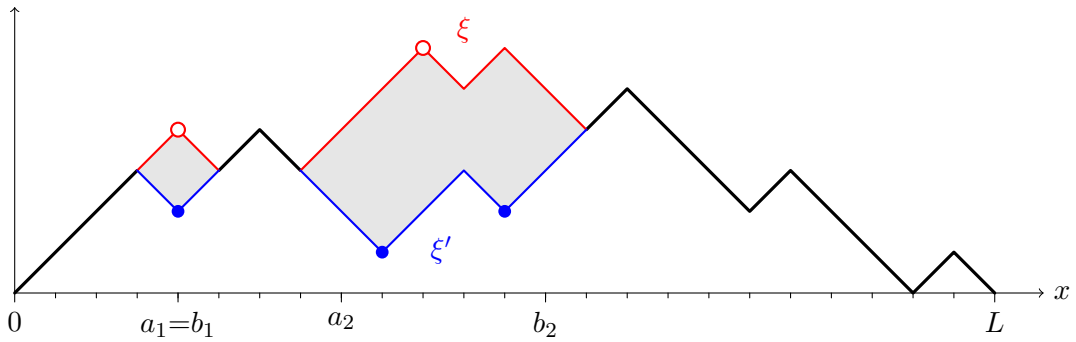


FIGURE 2. Two ordered paths $\xi \geq \xi'$ in Ω_{24}. The upper path ξ consists of the red and the thick black segments, the lower path ξ' of the blue and the thick black segments; the region between them is shaded. Here $S = I_1 \cup I_2$, with $I_1 = \{4\}$ and $I_2 = \{8, \dots, 13\}$. The open circles are the peaks $x_1 = 4$ and $x_2 = 10$ of ξ found in Step 3; their downward clocks move ξ only. The dots are the valleys of $\zeta = \xi'$ in S, at 4, 9 and 12, used in Step 4; their upward clocks move ξ' only. In I_2, these valleys cut the seven edges between 7 and 14 into the pieces $\{7, 8, 9\}$, $\{9, \dots, 12\}$ and $\{12, 13, 14\}$.

Moreover, $w_x \geq c_\beta$ for $1 \leq x \leq L-1$. Hence (3.6) gives

$$v(\xi, \xi') \geq \frac{4c_\beta^2}{1+\lambda} \, \#\mathcal{C}^*(\xi, \xi'). \tag{3.10}$$

It remains to exhibit sufficiently many active clocks. They will be attached to peaks of ξ and to valleys of ζ located in S.

Step 2: components of the discrepancy set. Since $\xi_x \equiv \xi'_x \equiv x \pmod 2$ and $\xi \geq \xi'$, the difference $\xi_x - \xi'_x$ is a nonnegative even integer. Hence

$$\xi_x \geq \xi'_x + 2 \geq 2 \qquad \text{for every } x \in S. \tag{3.11}$$

The set S is nonempty because $\xi \neq \xi'$, and $S \subset \{1, \dots, L-1\}$ because both paths vanish at 0 and L. Decompose S into its connected components, namely its maximal sets of consecutive integers:

$$S = I_1 \cup \cdots \cup I_p, \qquad I_k = \{a_k, a_k + 1, \dots, b_k\}, \qquad n_k = |I_k|,$$

where $p \geq 1$ and $b_k + 1 < a_{k+1}$ for $k < p$. By maximality, $a_k - 1$ and $b_k + 1$ are coincidence sites: they lie in $\{0, \dots, L\} \setminus S$. The heights ξ_{a_k} and ξ'_{a_k} differ by one from the common value $\xi_{a_k-1} = \xi'_{a_k-1}$, and $\xi_{a_k} \geq \xi'_{a_k} + 2$ by (3.11). Hence $\xi_{a_k} = \xi_{a_k-1} + 1$ and $\xi'_{a_k} = \xi'_{a_k-1} - 1$. The same argument at $b_k + 1$ gives

$$\begin{aligned} \xi_{a_k} - \xi_{a_k-1} &= 1, & \xi_{b_k+1} - \xi_{b_k} &= -1, \\ \xi'_{a_k} - \xi'_{a_k-1} &= -1, & \xi'_{b_k+1} - \xi'_{b_k} &= 1. \end{aligned} \tag{3.12}$$

Thus, between the coincidence sites $a_k - 1$ and $b_k + 1$, which are joined by $n_k + 1$ edges, the upper path starts with an ascending edge and ends with a descending one, while the lower path does the opposite; see Figure 2.

Step 3: a downward clock in each component. By (3.12), the set of $y \in I_k$ with $\xi_{y+1} - \xi_y = -1$ contains b_k. Let x_k be its smallest element. Then $\xi_{x_k} - \xi_{x_k-1} = 1$: for $x_k = a_k$ this is (3.12), and for $x_k > a_k$ it follows from the minimality of x_k, since then $x_k - 1 \in I_k$. Hence x_k is a peak of ξ. Its height $h_k = \xi_{x_k}$ is at least 2 by (3.11), so the label $c_k^\downarrow = (x_k, h_k, \downarrow)$ belongs to $\mathcal{C}$ and $\iota_{c_k^\downarrow}(\xi) = 1$. On the other hand, $\xi'_{x_k} \leq h_k - 2$, so $\iota_{c_k^\downarrow}(\xi') = 0$. Therefore $c_k^\downarrow \in \mathcal{C}^*(\xi, \xi')$: when this clock rings, the upper path flips its corner at x_k, while the lower path does not move.

Step 4: upward clocks at the valleys of ζ. Let V_k be the set of valleys of ζ contained in I_k, and let $m_k = \#V_k$. For $x \in V_k$, the label $c_x^\uparrow = (x, \zeta_x, \uparrow)$ belongs to $\mathcal{C}$ and $\iota_{c_x^\uparrow}(\zeta) = 1$. Since $x \in S$, we have $\tilde{\zeta}_x \neq \zeta_x$, so $\iota_{c_x^\uparrow}(\tilde{\zeta}) = 0$. Therefore $c_x^\uparrow \in \mathcal{C}^*(\xi, \xi')$.

We claim that

$$n_k + 1 \le 2Q(\zeta)(m_k + 1). \tag{3.13}$$

Write $V_k = \{y_1 < \cdots < y_{m_k}\}$, and put $y_0 = a_k - 1$ and $y_{m_k+1} = b_k + 1$. The *pieces* $\{y_i, y_i + 1, \dots, y_{i+1}\}$, $0 \le i \le m_k$, partition the $n_k + 1$ edges between $a_k - 1$ and $b_k + 1$. The interior sites of a piece lie in $I_k \setminus V_k$, so none of them is a valley of ζ. A descending edge immediately followed by an ascending edge would create a valley at their common site. Hence, along each piece, all ascending edges of ζ precede all descending ones, and the piece is made of at most two runs of consecutive edges with the same slope. Each run has at most $Q(\zeta)$ edges by (3.1), so each piece has at most $2Q(\zeta)$ edges. Summing over the $m_k + 1$ pieces proves (3.13).

Step 5: counting active clocks. The clocks $c_k^{\downarrow}$, $1 \le k \le p$, and $c_x^{\uparrow}$, $x \in V_1 \cup \cdots \cup V_p$, are pairwise distinct. Indeed, clocks obtained from different components are attached to different sites. Within I_k, the clock $c_k^{\downarrow}$ is a downward clock, whereas the clocks $c_x^{\uparrow}$, $x \in V_k$, are upward clocks attached to distinct sites. By Steps 3 and 4 and (3.13),

$$\#\mathcal{C}^*(\xi, \xi') \ge \sum_{k=1}^{p} (1 + m_k) \ge \sum_{k=1}^{p} \max\Big(1, \frac{n_k + 1}{2Q(\zeta)}\Big) \ge \max\Big(1, \frac{|S|}{2Q(\zeta)}\Big),$$

where the last inequality uses $p \ge 1$ and $n_1 + \cdots + n_p = |S|$. This proves the first inequality in (3.8).

Step 6: area and discrepancies. Since $0 < w_x \le 1$, $\xi' \ge 0$ and $\xi \le H$,

$$A = \sum_{x \in S} w_x(\xi_x - \xi'_x) \le \sum_{x \in S} \xi_x \le H|S|.$$

This is the second inequality in (3.8). Note that $H \ge 2$ by (3.11).

Step 7: conclusion. By (3.10), (3.8) and the elementary bound $\max(1, \frac{|S|}{2Q(\zeta)}) \ge \frac{1}{2}(1 + \frac{|S|}{2Q(\zeta)})$,

$$v(\xi, \xi') \ge \frac{4c_\beta^2}{1 + \lambda} \cdot \frac{1}{2}\Big(1 + \frac{|S|}{2Q(\zeta)}\Big) = \frac{c_\beta^2}{1 + \lambda}\Big(2 + \frac{|S|}{Q(\zeta)}\Big) \ge \frac{c_\beta^2}{1 + \lambda}\Big(1 + \frac{A}{HQ(\zeta)}\Big),$$

which is (3.9). Finally, by (3.5), the increment $A(T_c\xi, T_c\xi') - A(\xi, \xi')$ vanishes if $c \notin \mathcal{C}^*(\xi, \xi')$ and equals $\pm 2w_{x_c}$ otherwise. Since $c_\beta \le w_{x_c} \le 1$, every nonzero increment has absolute value in $[2c_\beta, 2]$. □

3.2. **Regularity at deterministic times.** The absorption estimate in Section 4 will use the expected time spent outside a set of regular configurations. It is therefore enough to control the probability of irregularity at each deterministic time.

Set

$$h_L = \sqrt{L} \log L, \qquad q_L = \lceil (\log L)^2 \rceil, \qquad K_L = h_L q_L, \tag{3.14}$$

and define

$$\mathcal{G}_t = \Big\{ \max_x X_t^+(x) \le h_L, \quad Q(Z_t) \le q_L \Big\}.$$

Lemma 3.2 (Typical height and monotone stretches)**.** *For each fixed $\alpha > 0$, if $T_L \ge (1 + \alpha)t_L$, then*

$$\sup_{t \ge T_L} \mathbb{P}(\mathcal{G}_t^c) \longrightarrow 0. \tag{3.15}$$

Proof. We first establish equilibrium tail bounds. Let S be simple symmetric random walk started at zero. Its law conditioned on $S \in \Omega_L$ is μ_L^1, and

$$\mathbb{P}(S \in \Omega_L) = 2^{-L} Z_L(1) \ge cL^{-3/2}$$

by Lemma 2.2. A union bound over positions and slope signs gives

$$\mu_L^1(Q \ge q) \le CL^{5/2} 2^{-q}, \qquad q \ge 1. \tag{3.16}$$

Also, the exponential martingale for S gives $\mathbb{P}(\max_{x\le L} S_x \ge h) \le \exp(-h^2/(2L))$ for $h > 0$. Indeed, for $a > 0$ the martingale $\exp(aS_x)/(\cosh a)^x$, stopped on reaching height h or time L, bounds this probability by $\exp(-ah + La^2/2)$; take $a = h/L$. Consequently,

$$\mu_L^1\Big(\max_x \xi_x \ge h\Big) \le CL^{3/2} \exp\left(-\frac{h^2}{2L}\right). \tag{3.17}$$

Choose $p > 1$ with $\lambda^p < 2$. For every event $D \subset \Omega_L$, Hölder's inequality and Lemma 2.2 give

$$\begin{aligned}\mu_L^\lambda(D) &= \frac{\mu_L^1(\lambda^{\mathcal{N}}\mathbf{1}_D)}{\mu_L^1(\lambda^{\mathcal{N}})} \\ &\le \left(\frac{Z_L(\lambda^p)}{Z_L(1)}\right)^{1/p} \mu_L^1(D)^{1-1/p} \le C_{\lambda,p}\mu_L^1(D)^{1-1/p},\end{aligned} \tag{3.18}$$

where the denominator in the first line is at least one. Applying (3.18) to (3.16) and (3.17), with $q = q_L$ and $h = h_L$, yields

$$\mu_L^\lambda(Q > q_L) + \mu_L^\lambda\Big(\max_x \xi_x > h_L\Big) = o(1).$$

Stationarity controls $Q(Z_t)$ uniformly in t.

For the upper path, put $\rho_L(t) = \|P_t^{\wedge,\lambda} - \mu_L^\lambda\|_{\mathrm{TV}}$. The extremal upper bound in [Yan21, Theorem 1.2] implies $\rho_L(T_L) \to 0$. Total variation contracts under the semigroup, so for every $t \ge T_L$,

$$\mathbb{P}\Big(\max_x X_t^+(x) > h_L\Big) \le \rho_L(T_L) + \mu_L^\lambda\Big(\max_x \xi_x > h_L\Big) = o(1).$$

Combining these two bounds proves the lemma. The supremum in (3.15) is outside the probability; the statement does not assert regularity at all times simultaneously. □

4. Coalescence and proof of the main theorem

4.1. **An absorption estimate.** We first give an absorption estimate for a nonnegative observable of a finite-state Markov chain. The function j below bounds the positive drift, the set G consists of states with regular geometry, and K is the scale appearing in the variance bound.

The guiding choice is a concave function with $-F''(a) = K/(K + a)$. This cancels the factor $1 + a/K$ in the variance lower bound and produces a negative drift of constant order while $A > 0$ and the state lies in G. A bound on F' then charges all positive drift to the expected integral of j, precisely the quantity controlled by Proposition 2.4.

Lemma 4.1 (Absorption with controlled positive drift)**.** *Let $(Y_t)_{t\ge0}$ be a continuous-time Markov chain on a finite set E, with transition rates $q(y, z)$ and generator $\mathcal{A}$. Let $A : E \to [0, B]$, where $B > 0$, and assume that $\{y : A(y) = 0\}$ is absorbing and that*

$$|A(z) - A(y)| \le 2 \qquad \text{whenever } q(y, z) > 0.$$

Define

$$b(y) = (\mathcal{A}A)(y), \qquad v(y) = \sum_{z\ne y} q(y, z)\big(A(z) - A(y)\big)^2.$$

Suppose that $b \le j$ for some function $j : E \to [0,\infty)$. Let $G \subset E$ and assume that, for some $K \ge 2$ and $c > 0$,

$$v(y) \ge c\left(1 + \frac{A(y)}{K}\right) \qquad \text{for all } y \in G \text{ with } A(y) > 0.$$

Set $A_t = A(Y_t)$ and $\ell = K\log(1+B/K)$. Then, for every initial distribution, every deterministic $T \ge 0$ and every $s > 0$,

$$\begin{aligned}\mathbb{P}(A_{T+s} > 0) \le \frac{4\ell}{cs}\left(\mathbb{E}A_T + \mathbb{E}\int_T^{T+s} j(Y_t)\,\mathrm{d}t\right) \\ + \frac{1}{s}\int_T^{T+s} \mathbb{P}(Y_t \notin G)\,\mathrm{d}t.\end{aligned} \tag{4.1}$$

Proof. Step 1: the function F. For $0 \le a \le B$, define

$$F(a) = K \int_0^a \log\left(\frac{K+B}{K+u}\right) \mathrm{d}u. \tag{4.2}$$

The right-hand side defines a smooth function of $a \in (-K, \infty)$, an interval containing $[0, B]$, and

$$F(0) = 0, \qquad F'(a) = K \log\left(\frac{K+B}{K+a}\right), \qquad F''(a) = -\frac{K}{K+a}.$$

For $a \in [0, B]$ we have $1 \le (K+B)/(K+a) \le 1+B/K$, hence $0 \le F'(a) \le K \log(1+B/K) = \ell$. Integrating this bound from 0 to a gives

$$0 \le F(a) \le \ell a, \qquad 0 \le F'(a) \le \ell, \qquad 0 \le a \le B. \tag{4.3}$$

Step 2: a second-order Taylor bound. We claim that

$$F(a') - F(a) - F'(a)(a'-a) \le -\frac{K(a'-a)^2}{4(K+a)} \qquad \text{if } a, a' \in [0, B] \text{ and } |a'-a| \le 2. \tag{4.4}$$

For $s \in [0,1]$ put $a_s = a + s(a'-a)$. Since $[0, B]$ is an interval, $a_s \in [0, B]$. Taylor's formula with integral remainder gives

$$F(a') - F(a) - F'(a)(a'-a) = (a'-a)^2 \int_0^1 (1-s) F''(a_s)\, \mathrm{d}s = -K(a'-a)^2 \int_0^1 \frac{1-s}{K+a_s}\, \mathrm{d}s.$$

For every $s \in [0,1]$,

$$0 < K + a_s \le K + a + |a'-a| \le K + a + 2 \le 2(K+a),$$

where the last inequality uses $K \ge 2$ and $a \ge 0$. Since $1 - s \ge 0$, it follows that

$$\int_0^1 \frac{1-s}{K+a_s}\, \mathrm{d}s \ge \frac{1}{K+a+2} \int_0^1 (1-s)\, \mathrm{d}s = \frac{1}{2(K+a+2)} \ge \frac{1}{4(K+a)},$$

which proves (4.4).

Step 3: the first inequality. Fix $y \in E$. The generator acts on functions $f : E \to \mathbb{R}$ by

$$\mathcal{A}f(y) = \sum_{z \ne y} q(y,z)(f(z) - f(y)).$$

In particular,

$$b(y) = \sum_{z \ne y} q(y,z)(A(z) - A(y)), \qquad \mathcal{A}(F \circ A)(y) = \sum_{z \ne y} q(y,z)(F(A(z)) - F(A(y))).$$

Let $z \ne y$ with $q(y,z) > 0$. Then $A(y), A(z) \in [0, B]$, and $|A(z) - A(y)| \le 2$ by assumption. Hence (4.4) applies with $a = A(y)$ and $a' = A(z)$, and gives

$$F(A(z)) - F(A(y)) \le F'(A(y))(A(z) - A(y)) - \frac{K}{4(K + A(y))}(A(z) - A(y))^2.$$

Multiply this inequality by $q(y,z)$ and sum over all such z; the terms with $q(y,z) = 0$ contribute nothing to $\mathcal{A}(F \circ A)(y)$, $b(y)$ or $v(y)$. The factors $F'(A(y))$ and $K/(4(K + A(y)))$ do not depend on z. Therefore the expressions above and the definition of v give

$$\mathcal{A}(F \circ A)(y) \le F'(A(y))\, b(y) - \frac{K}{4(K + A(y))}\, v(y). \tag{4.5}$$

Step 4: the second inequality. We bound the two terms on the right-hand side of (4.5) separately. For the drift term,

$$F'(A(y))\, b(y) \le F'(A(y))\, j(y) \le \ell j(y). \tag{4.6}$$

The first inequality holds because $F'(A(y)) \ge 0$ and $b(y) \le j(y)$; the second because $j(y) \ge 0$ and $F'(A(y)) \le \ell$, by (4.3). The nonnegative majorant j is needed here: if $b(y) < 0$ and $F'(A(y)) < \ell$, then $F'(A(y))b(y) > \ell b(y)$, so $F'(A(y))$ cannot be replaced by ℓ directly.

For the variance term, $v(y) \geq 0$, so the term is nonnegative. If moreover $y \in G$ and $A(y) > 0$, the assumed lower bound on v reads $v(y) \geq c(K + A(y))/K$, and therefore

$$\frac{K}{4(K + A(y))}\, v(y) \geq \frac{K}{4(K + A(y))} \cdot \frac{c\,(K + A(y))}{K} = \frac{c}{4}.$$

This is where the choice $-F''(a) = K/(K + a)$ enters: the factor $K/(K + A(y))$ exactly offsets the growth of the lower bound on $v(y)$ with $A(y)$. Combining the two cases,

$$\frac{K}{4(K + A(y))}\, v(y) \geq \frac{c}{4}\, \mathbf{1}_{\{A(y)>0\}} \mathbf{1}_G(y). \tag{4.7}$$

Inserting (4.6) and (4.7) into (4.5), we obtain, for every $y \in E$,

$$\begin{aligned} \mathcal{A}(F \circ A)(y) &\leq F'(A(y))\, b(y) - \frac{K}{4(K + A(y))}\, v(y) \\ &\leq \ell j(y) - \frac{c}{4}\, \mathbf{1}_{\{A(y)>0\}} \mathbf{1}_G(y). \end{aligned} \tag{4.8}$$

Step 5: Dynkin's formula. Since E is finite, Dynkin's formula applies to $F \circ A$: for every initial distribution, every deterministic $T \geq 0$ and every $s > 0$,

$$\mathbb{E}F(A_{T+s}) - \mathbb{E}F(A_T) = \mathbb{E} \int_T^{T+s} \mathcal{A}(F \circ A)(Y_t)\, \mathrm{d}t.$$

By (4.8), the right-hand side is at most

$$\ell\, \mathbb{E} \int_T^{T+s} j(Y_t)\, \mathrm{d}t - \frac{c}{4}\, \mathbb{E} \int_T^{T+s} \mathbf{1}_{\{A_t>0\}} \mathbf{1}_G(Y_t)\, \mathrm{d}t.$$

Rearranging, and using $F(A_{T+s}) \geq 0$ and $F(A_T) \leq \ell A_T$ from (4.3), we obtain

$$\frac{c}{4} \mathbb{E} \int_T^{T+s} \mathbf{1}_{\{A_t>0\}} \mathbf{1}_G(Y_t)\, \mathrm{d}t \leq \ell \left(\mathbb{E}A_T + \mathbb{E} \int_T^{T+s} j(Y_t)\, \mathrm{d}t \right). \tag{4.9}$$

Step 6: conclusion. Since $\{A = 0\}$ is absorbing, on the event $\{A_{T+s} > 0\}$ we have $A_t > 0$ for every $t \in [T, T + s]$. Hence, for every such t,

$$\mathbf{1}_{\{A_t>0\}} \mathbf{1}_G(Y_t) \geq \mathbf{1}_{\{A_{T+s}>0\}} (1 - \mathbf{1}_{\{Y_t \notin G\}}) \geq \mathbf{1}_{\{A_{T+s}>0\}} - \mathbf{1}_{\{Y_t \notin G\}}.$$

Taking expectations and integrating over $t \in [T, T + s]$, we see that the expected integral in (4.9) is at least

$$s\mathbb{P}(A_{T+s} > 0) - \int_T^{T+s} \mathbb{P}(Y_t \notin G)\, \mathrm{d}t.$$

Substituting this lower bound into (4.9) and dividing by $cs/4$ proves (4.1). □

4.2. **Proof of Theorem 1.1.**

Proof of Theorem 1.1. The lower bound in [Yan21, Proposition 3.1] states that, for some $C = C(\lambda, \varepsilon) < \infty$, $T_{\mathrm{mix}}^{L,\lambda}(\varepsilon) \geq t_L - CL^2$. In particular,

$$\liminf_{L\to\infty} \frac{\pi^2 T_{\mathrm{mix}}^{L,\lambda}(\varepsilon)}{L^2 \log L} \geq 1. \tag{4.10}$$

It therefore suffices to prove the upper bound.

Step 1: reduce the mean area below $L^{3/2}$. Fix $\beta \in (0, \pi)$ and $\eta \in (0, 1)$, independently of L, and set

$$T = \frac{1 + \eta}{\beta^2} L^2 \log L, \qquad s = L^2. \tag{4.11}$$

Consider either A^+ or A^-. Because $\gamma_\beta = \beta^2/(2L^2) + O(L^{-4})$, we have $\gamma_\beta T = \frac{1+\eta}{2} \log L + o(1)$, and Proposition 2.4 gives, for all sufficiently large L,

$$\mathbb{E}A_T \leq CL^{3/2-\eta/2}, \qquad \mathbb{E} \int_T^{T+s} j_t\, \mathrm{d}t \leq \frac{1 - \theta}{\theta} \mathbb{E}A_T. \tag{4.12}$$

Step 2: coalescence in an additional time L^2. Apply Lemma 4.1 to the finite-state chain $Y_t = (X_t^+, Z_t, X_t^-)$ on the invariant state space

$$E = \{(\xi, \zeta, \xi') \in \Omega_L^3 : \xi' \le \zeta \le \xi\},$$

with $A = A^+$ or $A = A^-$, $B = L^2$, $K = K_L \ge 2$ for all sufficiently large L, and

$$G = \left\{(\xi, \zeta, \xi') : \xi' \le \zeta \le \xi, \quad \max_x \xi_x \le h_L, \quad Q(\zeta) \le q_L\right\}.$$

Thus $\{Y_t \in G\} = \mathcal{G}_t$. The area is nonnegative and bounded by $L^2/4 \le B$, and zero is absorbing under the coupling. Its jumps are bounded by two by Lemma 3.1, and (2.6) gives $b_t \le j_t$. By (3.7), the quantity $v(Y_t)$ of Lemma 4.1 is v_t. For A^+, apply Lemma 3.1 with the reference path $\zeta = Z_t$. For A^-, apply it with the same reference path, now the upper member of the pair. In both cases, on $\mathcal{G}_t$ the upper member is bounded by X_t^+, so that $HQ(\zeta) \le K_L$ and

$$v_t \ge c_{\lambda,\beta}(1 + A_t/K_L) \qquad \text{whenever } A_t > 0.$$

The constants do not depend on L.

Since $K_L \asymp \sqrt{L}(\log L)^3$, we have

$$\ell_L := K_L \log(1 + L^2/K_L) \le C\sqrt{L}(\log L)^4.$$

Also $T \ge (1+\eta)L^2 \log L/\pi^2$, so Lemma 3.2 applies. Combining Lemma 4.1 with (4.12) therefore gives, for each sign,

$$\begin{aligned}\mathbb{P}(A^{\pm}_{T+L^2} > 0) &\le \frac{4\ell_L}{c_{\lambda,\beta}\theta L^2}\mathbb{E}A_T^{\pm} + \sup_{t\ge T}\mathbb{P}(\mathcal{G}_t^c)\\ &\le C_{\lambda,\beta,\eta}L^{-\eta/2}(\log L)^4 + o(1) \longrightarrow 0.\end{aligned}$$

All weights are strictly positive, so zero area is equivalent to equality of the two ordered paths. By (2.1),

$$d^{L,\lambda}(T + L^2) \longrightarrow 0.$$

Step 3: recover the sharp constant. For every fixed $\varepsilon \in (0,1)$ this proves

$$\limsup_{L\to\infty} \frac{T_{\mathrm{mix}}^{L,\lambda}(\varepsilon)}{L^2 \log L} \le \frac{1+\eta}{\beta^2}.$$

First let $\eta \downarrow 0$, then $\beta \uparrow \pi$. The resulting upper bound $1/\pi^2$, together with (4.10), proves the theorem. □

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